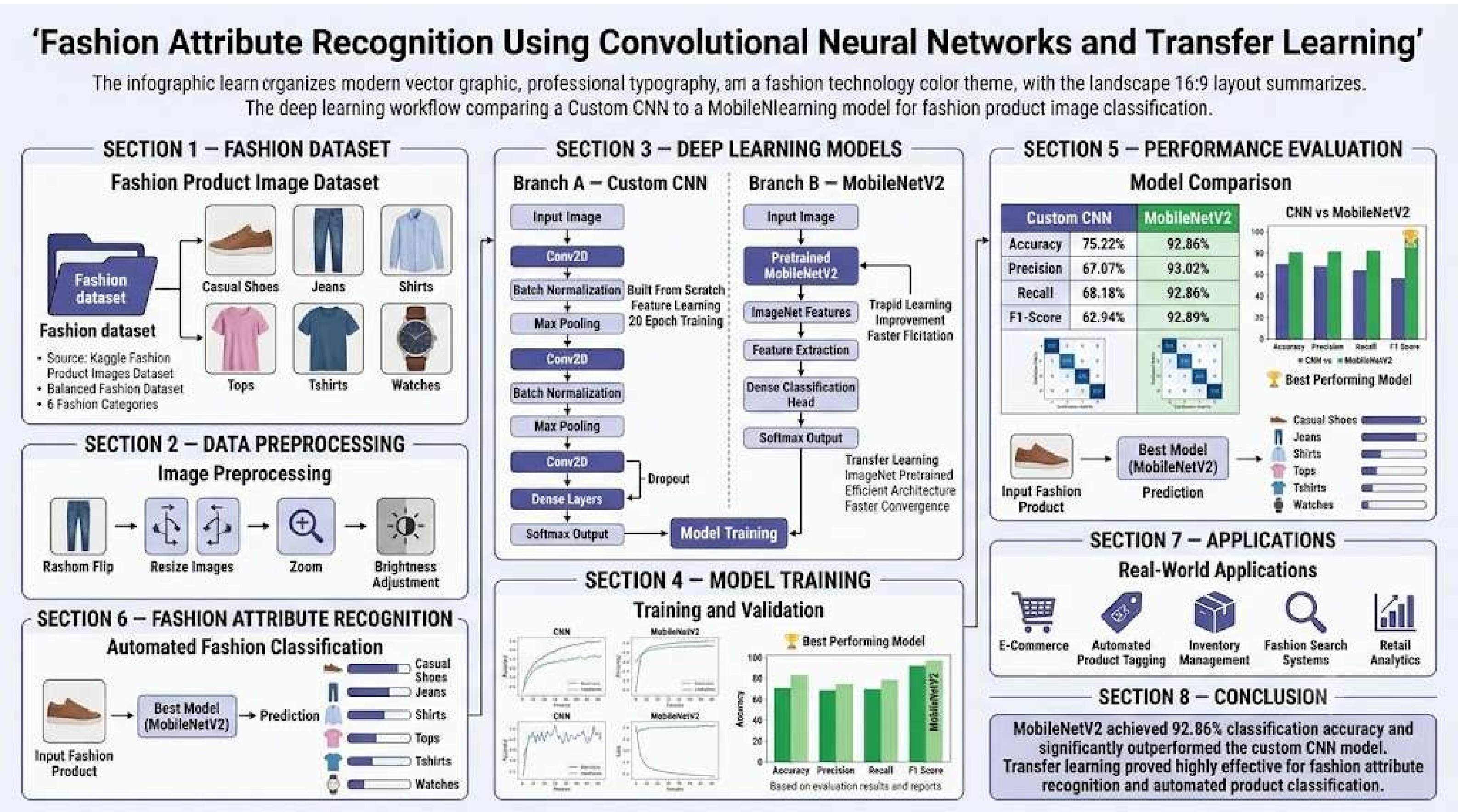
‘Fashion Attribute Recognition Using Convolutional Neural Networks and Transfer Learning’
The infographic learn organizes modern vector graphic, professional typography, am a fashion technology color theme, with the landscape 16:9 layout summarizes.
The deep learning workflow comparing a Custom CNN to a MobileNlearning model for fashion product image classification.
SECTION 1 — FASHION DATASET
Fashion Product Image Dataset
Fashion dataset
Casual Shoes
Jeans
Shirts
Tops
Tshirts
Watches
Fashion dataset
• Source: Kaggle Fashion Product Images Dataset
• Balanced Fashion Dataset
• 6 Fashion Categories
SECTION 2 — DATA PREPROCESSING
Image Preprocessing
Rashom Flip
Resize Images
Zoom
Brightness Adjustment
SECTION 6 — FASHION ATTRIBUTE RECOGNITION
Automated Fashion Classification
Input Fashion Product
Best Model (MobileNetV2)
Prediction
Casual Shoes
Jeans
Shirts
Tops
Tshirts
Watches
SECTION 3 — DEEP LEARNING MODELS
Branch A — Custom CNN
Input Image
Conv2D
Batch Normalization
Max Pooling
Conv2D
Batch Normalization
Max Pooling
Conv2D
Dense Layers
Softmax Output
Dropout
Built From Scratch
Feature Learning
20 Epoch Training
Model Training
Branch B — MobileNetV2
Input Image
Pretrained MobileNetV2
ImageNet Features
Feature Extraction
Dense Classification Head
Softmax Output
Trapid Learning Improvement Faster Ficitation
Transfer Learning
ImageNet Pretrained
Efficient Architecture
Faster Convergence
SECTION 4 — MODEL TRAINING
Training and Validation
CNN
MobileNetV2
CNN
MobileNetV2
Best Performing Model
Accuracy
Accuracy
Precision
Recall
F1 Score
MobileNetV2
Based on evaluation results and reports
SECTION 5 — PERFORMANCE EVALUATION
Model Comparison
Custom CNN
MobileNetV2
Accuracy 75.22% 92.86%
Precision 67.07% 93.02%
Recall 68.18% 92.86%
F1-Score 62.94% 92.89%
CNN vs MobileNetV2
Accuracy
Precision
Recall
F1 Score
Best Performing Model
Input Fashion Product
Best Model (MobileNetV2)
Prediction
Casual Shoes
Jeans
Shirts
Tops
Tshirts
Watches
SECTION 7 — APPLICATIONS
Real-World Applications
E-Commerce
Automated Product Tagging
Inventory Management
Fashion Search Systems
Retail Analytics
SECTION 8 — CONCLUSION
MobileNetV2 achieved 92.86% classification accuracy and significantly outperformed the custom CNN model.
Transfer learning proved highly effective for fashion attribute recognition and automated product classification.